

EE5311 Tutorial 3: CMOS Inverter Delay and Ring Oscillator

Ishan Aji Hassan
Roll No: EE23B097

All simulations use the SkyWater 130nm PDK (tt corner), $L_n = L_p = 0.15\mu m$, $W_n = 0.42\mu m$, run in ngspice via xschem.

Question 1: Static CMOS Inverter Delay

(a) Delay vs. W_p ($V_{DD} = 1.8V$)

The inverter drives an identical inverter (fanout of 1). Propagation delay $t_p = (t_{HL} + t_{LH})/2$ was measured using `.meas trig/targ` at the 0.9V ($V_{DD}/2$) crossing.

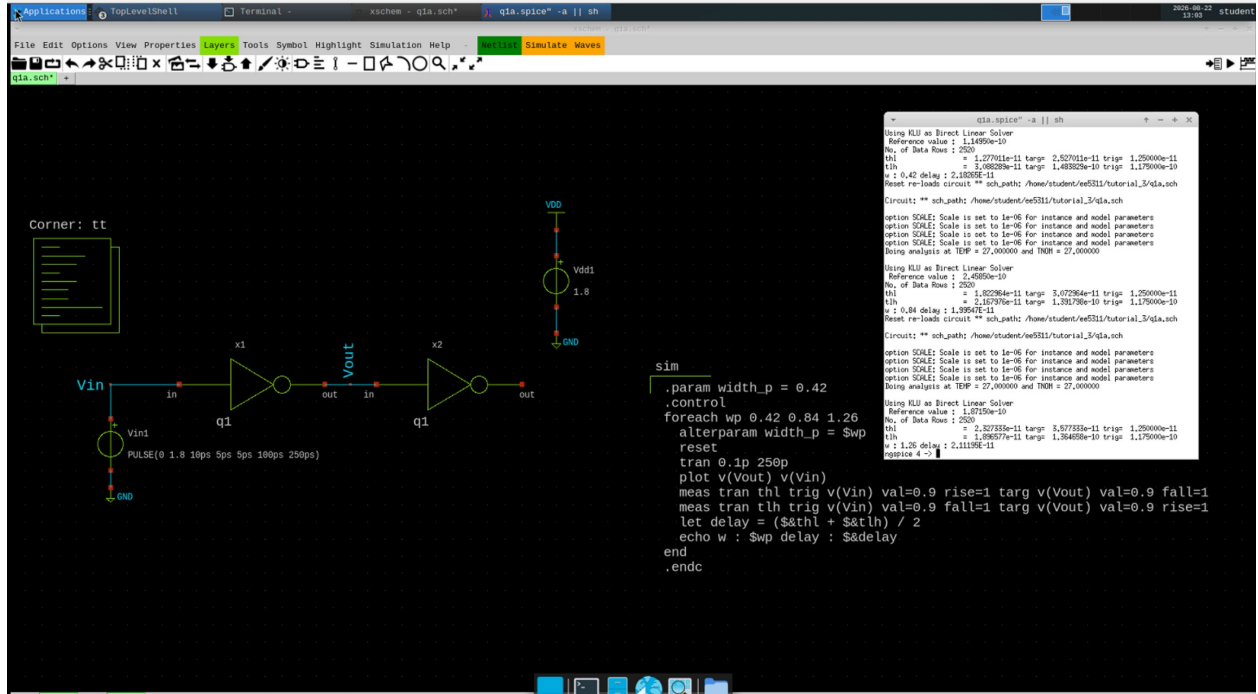


Figure 1: Schematic (two cascaded inverters, pulse input) and ngspice control code sweeping W_p .

W_p (μm)	t_p (ps)
0.42	21.83
0.84	19.95
1.26	21.12

Table 1: Propagation delay vs. PMOS width.

Minimum delay occurs at $W_p = 0.84\mu m$ ($W_p/W_n = 2$), which balances the pull-up/pull-down strength; this value is used for the rest of the assignment.

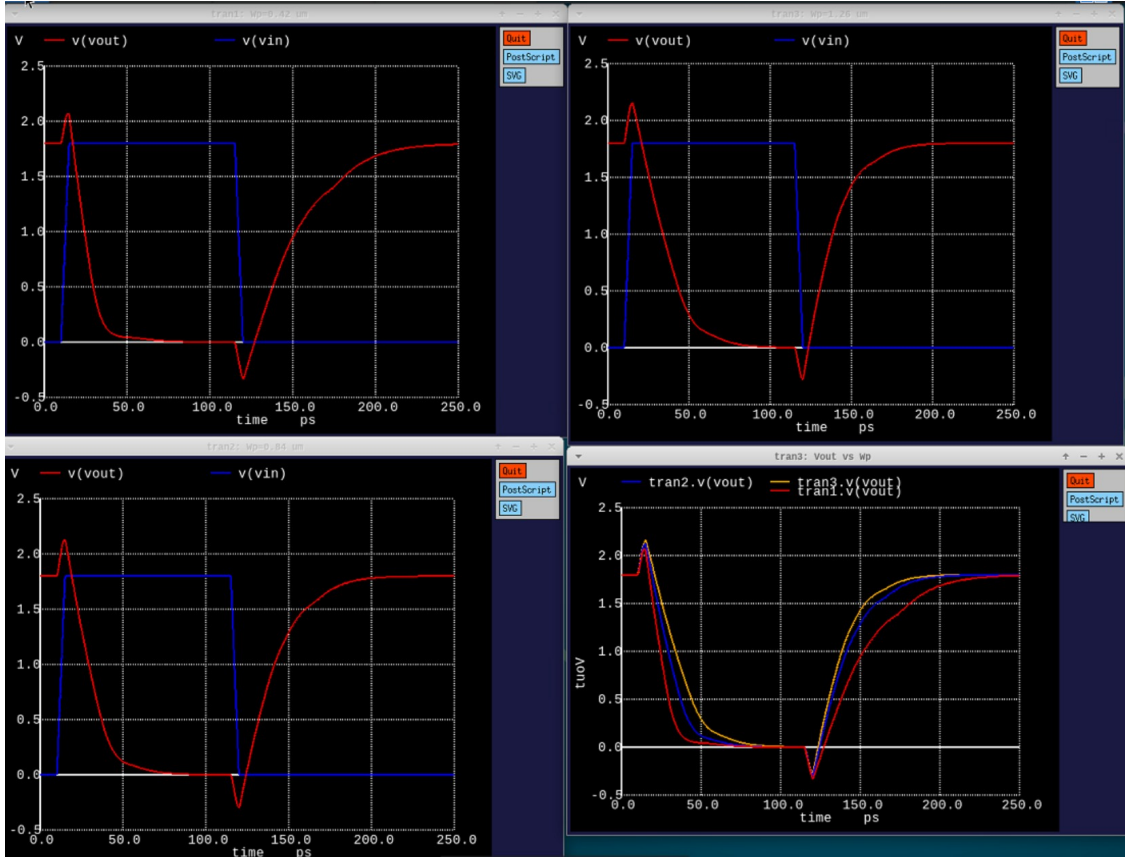


Figure 2: V_{out} and V_{in} transient waveforms for the three W_p values.



Figure 3: Overlay of V_{out} for $W_p = 0.42, 0.84, 1.26\mu m$ – the widest device settles fastest but the minimum t_p is at $W_p = 0.84\mu m$ due to input-slew/self-loading trade-off.

(b) and (c): t_p and Energy-Delay Product vs. V_{DD}

Using $W_p = 0.84\mu m$, V_{DD} was swept from 1.0V to 1.8V. Delay and EDP were measured for each point using the `alterparam/reset` loop in ngspice.

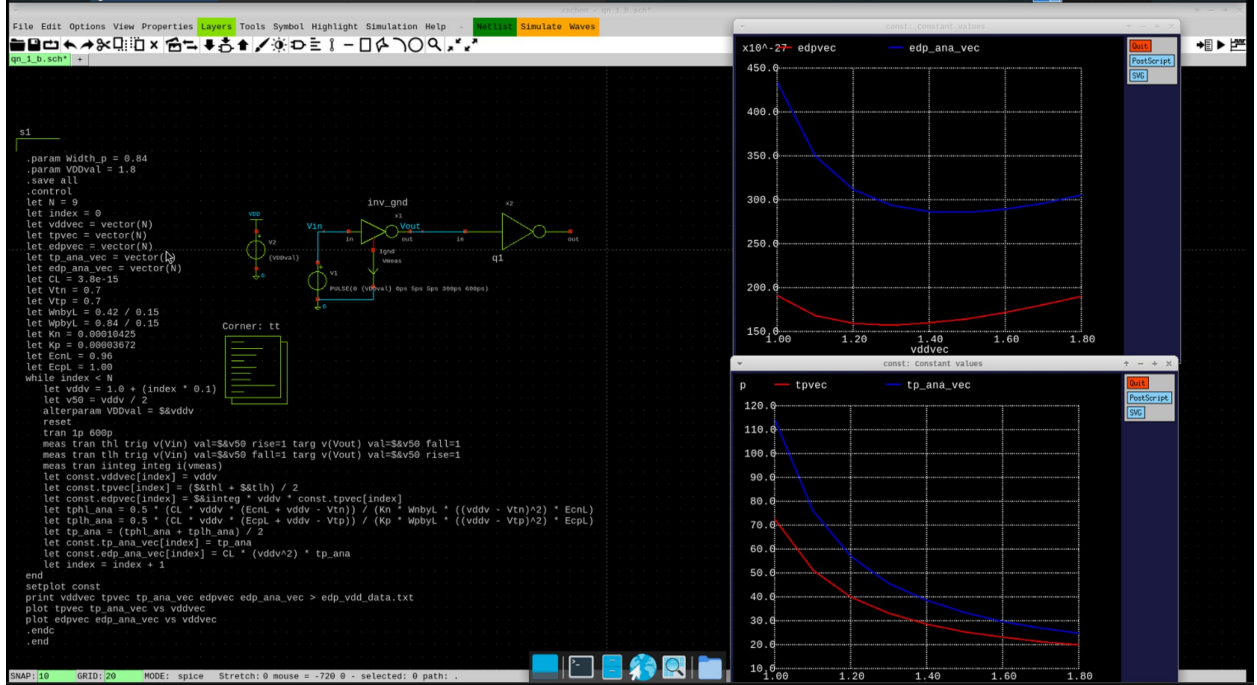


Figure 4: Left: propagation delay vs. V_{DD} (decreases from $\sim 65ps$ at 1.0V to $\sim 20ps$ at 1.8V). Right: energy-delay product vs. V_{DD} ($\times 10^{-27}J\cdot s$), monotonically increasing with V_{DD} .

Delay decreases with V_{DD} since drive current increases faster than load capacitance/voltage

swing, consistent with the analytical estimate $t_p \propto \frac{C_L V_{DD}}{I_{DSAT}}$. EDP increases monotonically over this range, so it has no interior minimum here – indicating the optimum V_{DD} for minimum EDP lies below 1.0V, outside the swept range; within the given 1.0–1.8V range, lower V_{DD} is more energy-delay efficient.

Question 2: Seven-Stage Ring Oscillator

Built from the minimum-delay inverter of Q1(a) ($W_p = 0.84\mu m$), 7 stages in a loop, with `.ic v(Vout)=0` and `uic` to force startup oscillation.

(a) Oscillation frequency at $V_{DD} = 1.8V$

Period	Frequency
409.1 ps	2.444 GHz

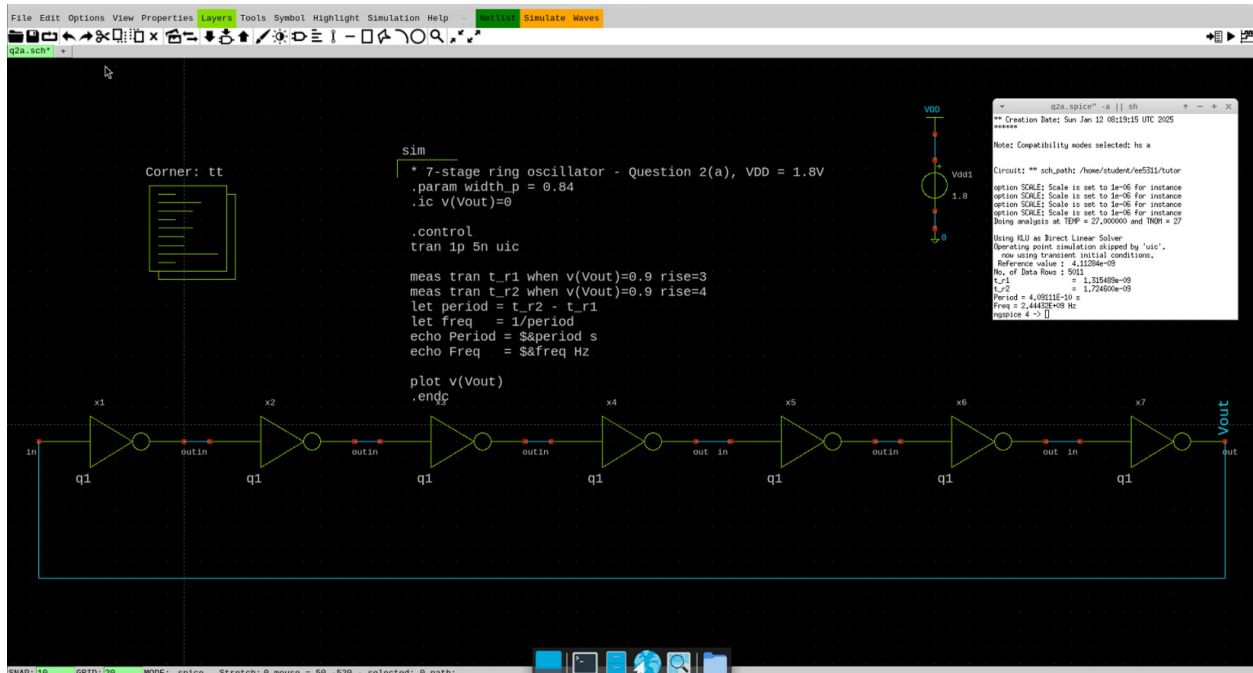


Figure 5: 7-stage ring schematic and ngspice control code (`.ic v(Vout)=0`, `uic`, period/frequency measurement).

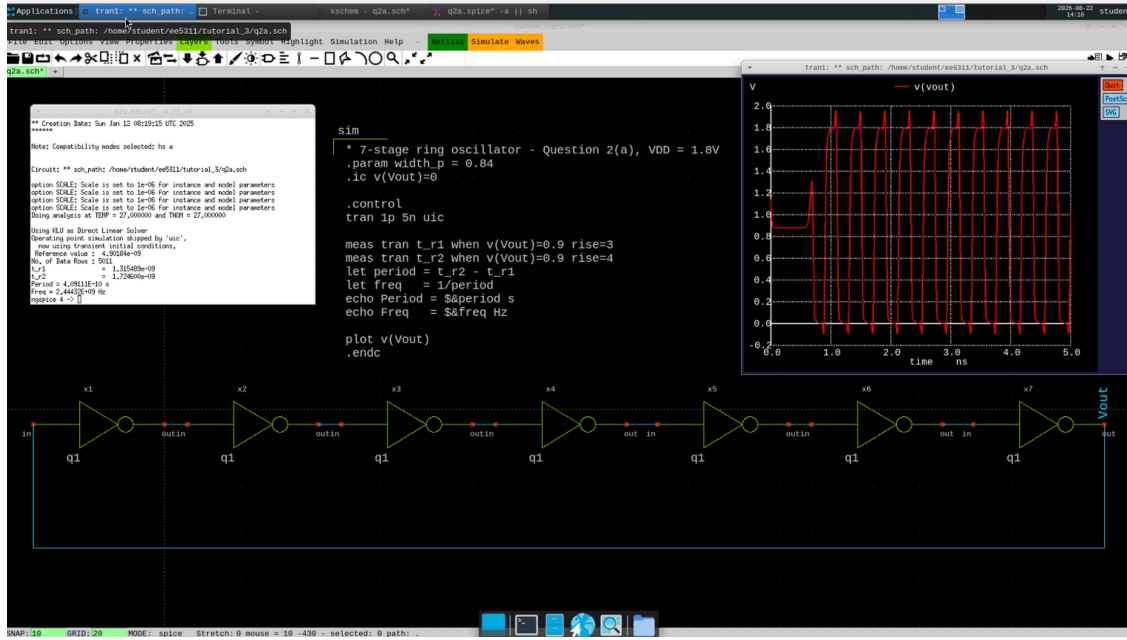


Figure 6: V_{out} showing startup transient followed by steady periodic oscillation.

(b) Frequency and period vs. V_{DD} (7 stages)

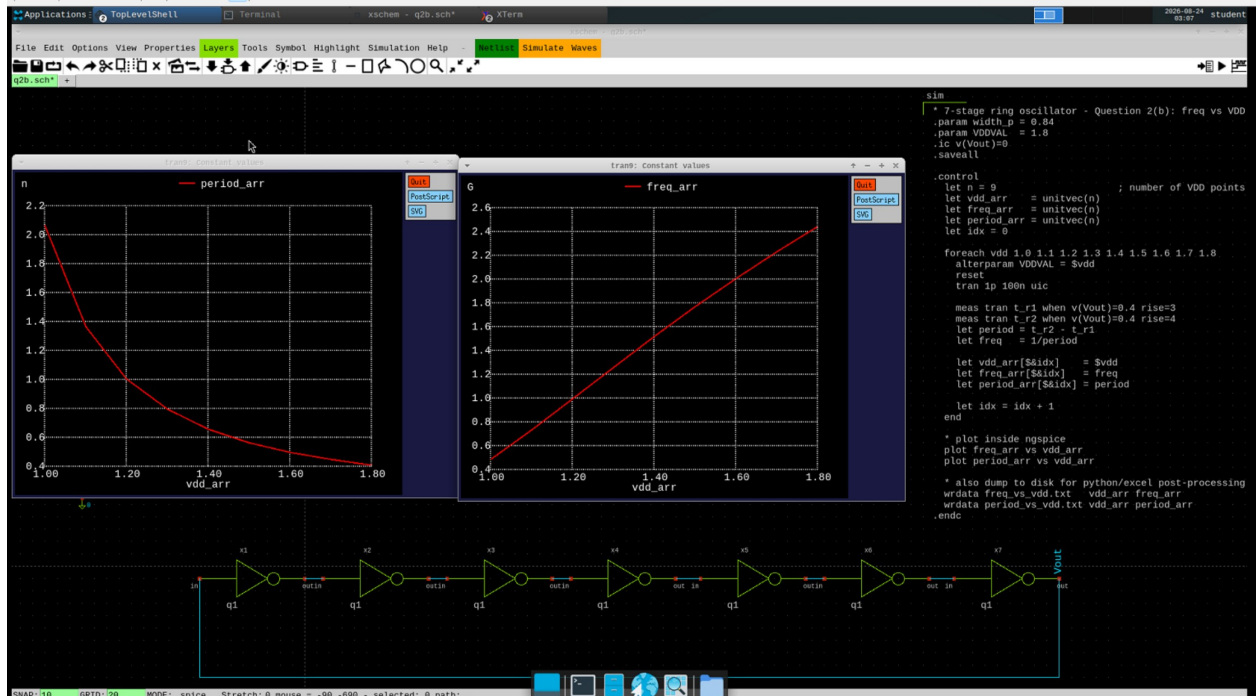


Figure 7: 7-stage ring schematic, ngspice control code (foreach V_{DD} sweep with dataset-based tranN.period/tranN.freq summary), and console output.

V_{DD} (V)	Period (ps)	Frequency (GHz)
1.3	794.4	1.259
1.4	659.2	1.517
1.5	566.0	1.767
1.6	498.7	2.005
1.7	448.1	2.232
1.8	409.1	2.444

Table 2: Below $V_{DD} = 1.3\text{V}$ the ring did not complete 4 clean oscillation cycles within the simulated 10ns window (weak drive at low overdrive), so no result is reported for 1.0–1.2V.

Frequency increases with V_{DD} since per-stage delay t_p decreases with V_{DD} (Q1b), and $f \approx \frac{1}{2Nt_p}$ for N stages.

(c) Repeat with nine stages

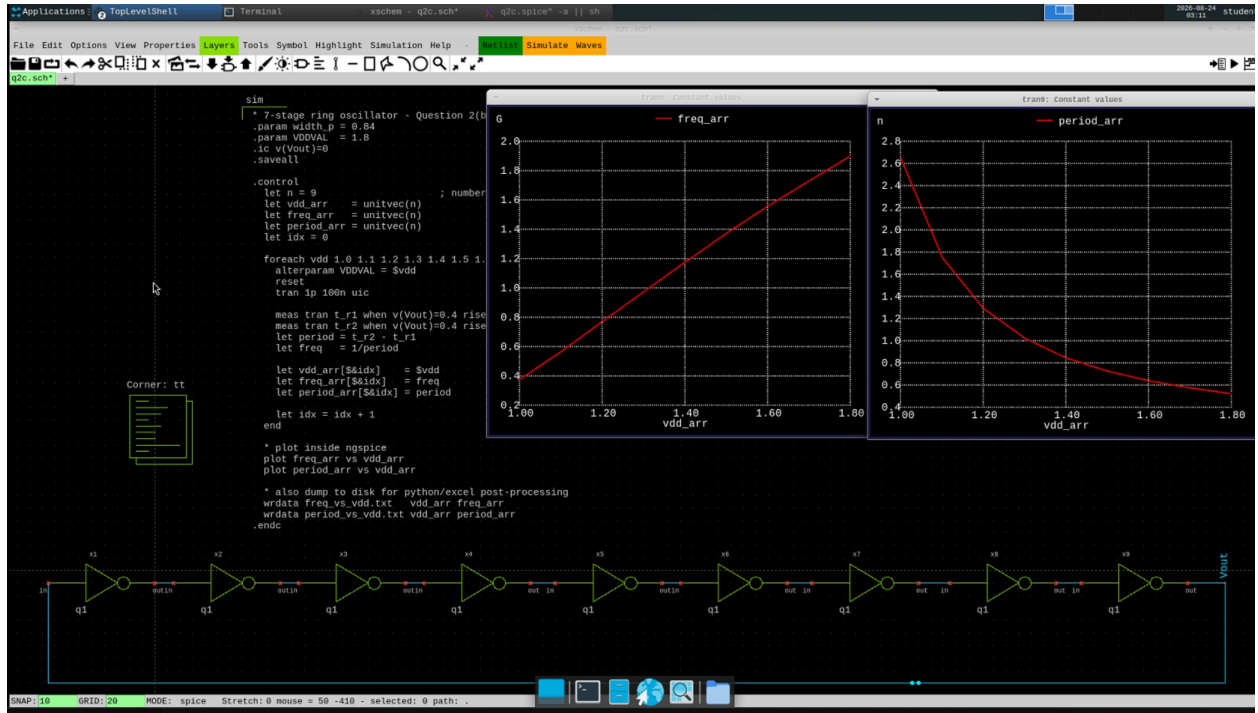


Figure 8: 9-stage ring schematic, same control code as (b), and console output.

V_{DD} (V)	Period (ps)	Frequency (GHz)
1.3	1021.4	0.979
1.4	847.5	1.180
1.5	727.7	1.374
1.6	641.2	1.560
1.7	576.2	1.736
1.8	526.0	1.901

Table 3: Same behavior as the 7-stage case; again no result below 1.3V for the same reason.

With 9 stages the period is longer and frequency lower than the 7-stage case at every V_{DD} , consistent with $f \propto 1/N$: more stages means a longer loop delay per oscillation cycle.